

Each normalization technique serves different purposes:

- **Min-Max Scaling** is useful when you need a specific range, like inputs to certain machine learning algorithms.
- **Standardization** is valuable for comparing volatility and relative movements.
- **Percent Change** is most intuitive for comparing performance across assets with different price levels.

?? Did You Know? During the financial crisis of 2008, many quantitative hedge funds suffered devastating losses in what became known as the "Quant Quake." A key factor in this meltdown was that their models failed to account for the changing correlations and volatility in their normalized data. When market behavior suddenly changed, their standardization assumptions broke down. The surviving firms, like D.E. Shaw and AQR Capital, subsequently improved their normalization techniques to dynamically adjust to changing market conditions rather than using static historical parameters. Today, these firms employ adaptive normalization methods that continuously update based on recent market behavior.

Adjusting for Corporate Actions: Splits and Dividends

Stock splits and dividend payments can create artificial price jumps or drops that distort analysis. Most data providers offer adjusted prices that account for these events, but understanding the adjustments is important:

```
1. import yfinance as yf
2.
3. # Download both adjusted and unadjusted data
4. data = yf.download("TSLA", start="2020-01-01", end="2023-07-01",
5.                    auto_adjust=False) # Get unadjusted data plus adjustment factors
6.
7. # Display both raw close price and adjusted close
8. comparison = data[['Close', 'Adj Close']].tail(10)
```